

Scenario Development for Decision-Making in Water Resources Planning and Management

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Abstract—Scenario planning is a useful approach to deal with high levels of uncertainty when planning, developing, and managing water resources systems. Scenarios should highlight many possible alternatives in the future and each alternative future is not equally likely. Probabilistic scenarios are proposed. An example is given to demonstrate the process to generate the scenarios with probabilities, which indicates a promising new insight the technique may provide. Scenarios are able to inform the decision making process, conversely, be applied to test the robustness of strategies. Therefore, an interactive scenario-decision system can be built to support decision making process in water resource planning and management.

Key words—uncertainty, scenario development, probabilistic scenarios, interactive scenario-decision making

1 Introduction

Uncertainty refers to the situation in which no unique and complete understanding of the system to be managed exists (Brugnach et al., 2009). Uncertainty is always present when planning, developing, and managing water resources systems. The uncertainty is usually generated by unpredictable or incomplete knowledge, such as the knowledge about future meteorological, demographic, economic, social, technical and political conditions which would affect the performance of water resources systems. On the other hand, the diverse discipline background of decision makers and the communication barrier between decision-makers and stakeholders would also produce uncertain situations when making decisions.

Scenario planning is a useful approach when dealing with high levels of uncertainty about water resources systems. Walters mentioned that scenario planning is somewhat similar to adaptive management, which takes uncertainty into account (Walters, 1986). Cornish mentioned that scenarios have a primary ability to solve almost all the important but uncertain situations effectively (Cornish, 2005). Scenario-based approaches have been applied to predict and analyze the future water issues, also support water managers and policy-makers to put forward solutions to deal with the potential problems (Middelkoop et al., 2002; Shiklonamov and Balonishnikova, 2003; Katsiardi et al., 2005b; Katsiardi et al., 2005a; Pallottino et al., 2005; Lienert et al., 2006; Wen et al., 2007; Sharda et al., 2009; Mahmoud et al., 2010; Weng et al., 2010). One notable example is the scenarios development about future

water supply and demand situations in a 50-year time horizon for the Verde River Watershed in Arizona, USA. Three dimensions including climate change (periodic droughts vs. sustained droughts), demographics (water-conservative population vs. water-consumptive population) and economy (booming economy vs. poor economy) were identified to form eight scenarios by different combinations. The scenarios analysis results point out that the augmentation of water supplies by out-of-basin water transfer is not a feasible long-term solution, while the only way to truly maintain the local water resources is to change water consumption habits and behavioral patterns. This is what the water managers and policy-makers must achieve before the water sustainability decreases (Mahmoud et al., 2010).

Scenarios can help us take a long view in a world of great uncertainties; additionally, the scenario analysis outcomes can enhance decision-making activities. Although scenarios do not provide precise knowledge about what will happen in the future, they contribute to better understanding of alternative futures states. More importantly, scenarios provide informative references for decision-makers and stakeholders who are required to take actions to adapt to the possible futures. The opposite is also true: decision-makers can rank multiple alternative decisions by scenario testing so as to examine the robustness of the strategies or decisions. Therefore, an interactive scenario-decision making system could be useful in water resource planning and management (Fig.1).

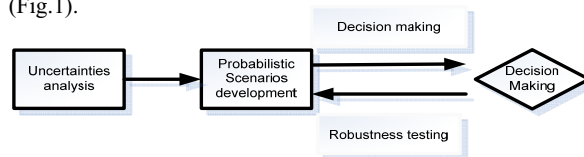


Fig. 1 An interactive scenario-decision making under uncertainties

2 What are scenarios?

Scenarios have been defined in several ways, but the common kernel of them is concerning the power of scenarios to think of futures in a consistent and plausible manner. Porter (1985) defined scenario as “an internally consistent view of what the future might turn to be - not a forecast, but one possible future outcome” (Porter, 1985). Schwartz defined scenario as “a tool for ordering one’s perception about alternative future environments in which one’s decisions

might be played out” (Schwartz 1991b). The IPCC (Intergovernmental Panel of Climate Change) defined scenario as “a coherent, internally consistent and plausible description of a possible future state of the world. It is not a forecast; rather, each scenario is one alternative image of how the future can unfold.” (http://www.ipcc-data.org/ddc_definitions.html). Although it is difficult to find a crisp definition for scenarios, scenarios are obviously not equal to forecasts or predictions of the future. Unlike forecasts or predictions, scenarios aim at encouraging people to think about the full picture but do not only focus on getting the future ‘right’. Besides, scenarios should not only focus on the best case, middle case or worst case. A broad view of plausible alternative futures should be constructed by exploring various scenarios. In this way, it is helpful to stress the irreducible uncertainties that are beyond the control of the people making decisions (Peterson et al., 2003).

3 Overview of scenario planning techniques

Scenarios are primarily described qualitatively and rely strongly on the personal judgment of scenario developers. Genius forecasting was first introduced by Herman Kahn, the original scenerist, who encouraged people to think the unthinkable. Markley and Herman in the 1970s employed relaxation and meditative techniques to relax the analytical mind, and then produced and visualized more intuitive images of the future (Markley, 1988). The trend extrapolation technique produces one elaborated baseline scenario, as the expected future, using judgment or mathematical techniques, but Herman Kahn claimed in his often-quoted phrase, “The most likely future isn’t.” Multiple scenarios were explicitly considered to give alternative futures after Kahn’s genius judgment and trend exploration. The SRI matrix, developed by Stanford Research Institute and used by Hawken in his book *Seven Tomorrows* (Hawken et al., 1982), identifies scenario categories and elements in two dimensions of the matrix. Participants then fill in the domain with the state of each element in the scenario categories. The GBN (Global Business Network) matrix, based on two dimensions of uncertainty or polarities, contains a kernel scenario in each cell. It has become the default scenario technique since the best-seller book of Schwartz (1991), *The Art of the Long View* (Schwartz 1991a) was published. Morphological Analysis (MA) or field anomaly relaxation (FAR) is an extension of the GBN, which contain any number of uncertainties and alternative states of uncertainty (Coyle et al., 1994; Ritchey, 2007-2009; Yang et al., 2010).

Scenarios planners do not only develop techniques to explore what can happen in the future, but are also curious about how likely it will happen. In 1966, two colleagues of Herman Kahn generated a quasi-quantitative approach based on the cross-impact analysis, that is, to analyze the inter-related impacts between a set of forecasts. One colleague, Theodore Gordon, who founded the Futures Group in Connecticut, developed and ameliorated the trend impact analysis. Trend extrapolation based on the historical data is used to construct a baseline scenario, and whether future events are believed important is specified by their probabilities and impacts on the baseline scenario (Gordon and Hayward, 1968; Gordon, 1994). Olaf Helmer, with his assistant Selwyn

Enzer, invented a computer-based cross-impact model to generate multiple scenarios, a cross-impact matrix established first by expert judgment and then run many times by a computer to evaluate the conditional probabilities between the events and the final probabilities. Cross-impact analysis grew into a family of techniques for future research. SMIC74, based on cross-impact analysis, adjusts the conditional probability given by experts to conform to the laws of probability $P(i)=P(i/j) \cdot P(j)+P(i/\bar{j}) \cdot P(\bar{j})$ (Duperrin and Godet 1975). MORPHOL is a computer program to manage the complexity of Morphological Analysis by reducing the number of uncertainties and their combination. The probability of each scenario can be calculated and compared with the user-defined mean probability of all scenarios (Godet and Roubelat, 1996). Probability trees (Lisewski, 2002) graphically branch the future events orderly and assign probabilities to each branch, hence, the probabilities of final states of events can be calculated. Table 1 summarizes the overview of scenario planning techniques.

Tab.1 Overview of scenario planning techniques

Basis	Technique	Process	Outcomes
Judgment	Genius forecasting	Thinking, imagining	One or more scenarios
	Visualization	Relaxation and Intuitive imagination	One or more scenarios
	Baseline	Trend extrapolation and elaboration	Elaborated baseline scenario
	SRI	Create 2-dimension matrix with scenario categories and elements, specify and elaborate the matrix	Multiple scenarios in specific domains
	GBN	Create 2*2 matrix by two most important and most uncertain, title and elaborate	Four mutually exclusive scenarios
	Morphological analysis	Extension of GBN with multiple dimensions	Multiple scenarios
Quantification	Trend impact analysis	Estimate probabilities of events or their impact on baseline	Adjusted trends
	Cross-impact analysis	Assign initial probability of each event, conditional probabilities of each other	Final probabilities of each event of each state
	SMIC74	Adjust the conditional probabilities based on cross-impact analysis	Final probabilities of each event of each state
	MORPHOL	Perform morphological analysis by computer program	Probability of each alternative
	Probability trees	Branch uncertainty or choice points sequentially and assign probabilities	Probability of each state

Source: (Bishop et al., 2007)

4 Scenarios with probabilities

Not all scenarios are equally likely; as a result, developing scenarios with probabilities are more reasonable and essential. Schneider (2001) mentioned that the absence of any probability assignment would lead to ambiguity and potential dangers, as users select scenarios intuitively or assume equal likelihood (Schneider, 2001). David Kline, supervisor of the gas forecasting and model development unit of the California

Energy Commission, claimed that to bridge the gap between scenarios and probability would be the most important intellectual and practical endeavor of our time.

One immediate challenge for scenario developers in applying probability methods is how to handle incompleteness of scenarios, because the scenarios developed may not fully include all possible futures. The IPCC Fourth Assessment Report pointed out two problems with climate-model-based scenarios: firstly, limited number of scenarios is not sufficient for adaptive planning because of the large range of different climate-model-based scenarios (Prudhomme et al., 2003; Nawaz and Adeloye, 2006). Secondly, it is difficult to assign a specific probability to an individual scenario, but it is possible to assign probability distributions of the different drivers of climate change by making assumptions (IPCC(AR4), 2007). Thus, probability distribution can be used to handle both completeness and inequality of scenarios.

5 An example of probabilistic scenarios

This example demonstrates the generation of probabilistic scenarios and their advantages. θ_1 and θ_2 are two variables which should be taken into account to develop scenarios. In order to cover all the possible future changes of the variables, the future states of the two variables are continuous within the upper and lower values. As the values of the two variables are not equally likely, normal distribution is assigned to the variables. The values and the corresponded pdf (probability density function) are as follows:

$$\theta_1 \sim N(\theta_1, 4, 1), \theta_1 \in (-2, 10);$$

$$\theta_2 \sim N(\theta_2, -5, 1), \theta_2 \in (-15, 5).$$

When computing the probabilistic scenarios consisting of the two variables, the joint probability is going to be used. In reality, two variables can be either uncorrelated or correlated. Hence, both two cases are considered.

(1). When θ_1 and θ_2 are uncorrelated, the joint probability of the two variables is calculated as the equation 1:

$$P(\theta_1, \theta_2 \in \theta) = \iint_{\theta} f(\theta_2)f(\theta_1)d\theta_1d\theta_2; \quad (1)$$

Where $P(\theta_1, \theta_2 \in \theta)$ is the joint probability of θ_1 and θ_2 when the variables fall inside the domain θ . θ is located in the 2-dimensional space of the values of θ_1 and θ_2 .

(2). When θ_1 and θ_2 are correlated, a bivariate normal distribution is formed by the two normally distributed variables and the joint probability distribution function is shown in the equation 2:

$$f(\theta_1, \theta_2) = \frac{1}{2\pi\sigma_{\theta_1}\sigma_{\theta_2}\sqrt{1-\rho_{\theta_1, \theta_2}^2}} \exp\left(-\frac{1}{2(1-\rho_{\theta_1, \theta_2}^2)} \left[\frac{(\theta_1 - \mu_{\theta_1})^2}{\sigma_{\theta_1}^2} + \frac{(\theta_2 - \mu_{\theta_2})^2}{\sigma_{\theta_2}^2} - \frac{2\rho_{\theta_1, \theta_2}(\theta_1 - \mu_{\theta_1})(\theta_2 - \mu_{\theta_2})}{\sigma_{\theta_1}\sigma_{\theta_2}} \right] \right) \quad (2)$$

Where μ_{θ_1} : the expected value of θ_1 ; μ_{θ_2} : the expected value of θ_2 ; σ_{θ_1} : the standard deviation of θ_1 ; σ_{θ_2} : the standard deviation of θ_2 . $\rho_{\theta_1, \theta_2}$ is the correlation coefficient of θ_1 and θ_2 .

Normally, the correlation coefficient of the two variables $\rho_{\theta_1, \theta_2}$ can be represented by the Population Pearson correlation coefficient and calculated based on equation 3:

$$\rho_{\theta_1, \theta_2} = \text{corr}(\theta_1, \theta_2) = \frac{\text{cov}(\theta_1, \theta_2)}{\sigma_{\theta_1}\sigma_{\theta_2}} = \frac{E[(\theta_1 - \mu_{\theta_1})(\theta_2 - \mu_{\theta_2})]}{\sigma_{\theta_1}\sigma_{\theta_2}} \quad (3)$$

It is assumed that the correlation coefficient is 0.5.

Fig.1 and Fig.2 provides a complete view of all the probabilistic scenarios combined by the two variables. They include the most likely cases, extreme cases and the cases between the extremes, which display that a characteristic of scenarios is to cover all the possible future states. More importantly, the probability corresponding with each scenario transmits information on the likelihood of alternative futures, which can enhance the decision-making process. Additionally, whether the variables are correlated or not has a large influence on the final probability of the scenario, so it is critical to analyze the interrelation of the variables.

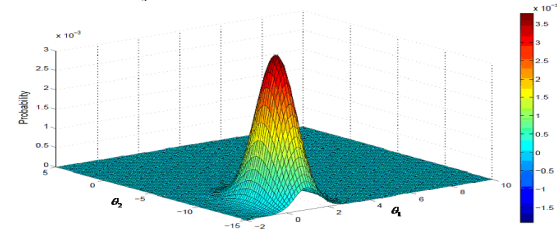


Fig.2 Probabilistic scenarios when two variables are uncorrelated

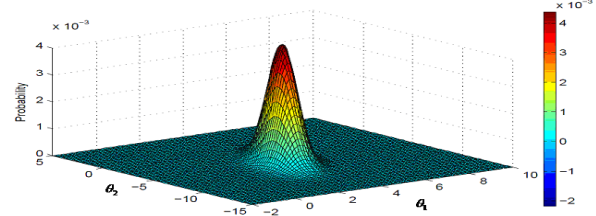


Fig.3 Probabilistic scenarios when two variables are correlated

6 Conclusions

Scenarios development should not only be explored as “the most likely futures”. Scenarios will be constrained to show a limited future when only best case, middle case or worst case is considered. However, the advantage of scenario development is to inform the decision-makers about alternative futures instead of the most-likely ones. A full spectrum of possible futures can provide more information for decision-makers.

Probabilistic scenarios can help decision-makers to test the robustness of strategies and decisions. However, it is complicated to generate probabilistic scenarios because of the need to quantify the future trends of variables, the interrelation

between them and also the corresponding probability. The example given in the paper demonstrated the potential of probabilistic scenarios, although so far only two variables were involved.

Scenarios have already been adopted frequently in water resource management studies when futures are needed to be considered, while scenarios applications related to water resources in most of the studies actually do not emphasize the scenario theory or scenario development approach, but the practical utility of scenarios as input to water management process. The problem to apply scenario science to water management is the lack of a scenario development framework. More future research should be devoted to building a systematic and interactive scenario developing and decision making framework in water resources management.

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